S5 has the finite model property

The Open Logic Project contributors

openlogic:normal-modal-logic/filtrations:local-snapshot-20261013;
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Editorial note. Fix note (Chief dispatch 2026-10-14 22:1x, batch 199 closeout): the stub number field of external reference prop:S5=univ contained the equals sign inherited from the source label; the number field is the card-face stub naming layer, not the external label layer, so it was changed to prop-S5-univ while the label field keeps the true source label prop:S5=univ. Same repair already applied to the sibling card X2-normal-modal-logic-filtrations-univ-fin; no other field touched. Revision bumped r001->r002 after the first assembly attempt was rejected by pd with original-reference-number-required.

Editorial scope. One original corollary of the Open Logic Project unit normal-modal-logic/filtrations, counted once: S5 has the finite model property. Only the counted corollary and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- S5 frames as reflexive and euclidean models, and the frame-definability unit's theorem that a formula true at a world in a reflexive and euclidean model is true at a world in a universal model.
- The unit proposition that universal validity is decided on finite universal models (prop:univ-fin of this unit).
- Every universal model is reflexive and euclidean.

Source reference. Original proposition prop-S5-univ: sources/olp-normal-modal-logic-frame-definability/pinned-original-source.tex, source lines 657–657. The author proof cites the frame-definability unit's universal-model characterization to move from a reflexive and euclidean model to a universal one; the defining unit is a sibling OLP unit, so the reference records the label in that unit's pinned source.

Source reference. Original proposition prop-univ-fin: sources/olp-normal-modal-logic-filtrations/pinned-original-source.tex, source lines 482–482. The author proof cites this unit's finite-universal proposition, which also supplies the explicit filtration witness; the counted corollary retains the final observation that universal models are reflexive and euclidean.

Corollary 0.1. *S5 has the finite model property.*

Proof. By prop-S5-univ , if A is true at a world in some reflexive and euclidean model then it is true at a world in a universal model. By prop-univ-fin , it is true at a world in a finite universal model (namely the filtration of the model through the set of sub-formulas of A). Every universal model is also reflexive and euclidean; so A is true at a world in a finite reflexive euclidean model. □